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DAY 3

Task 1

Human Pose Detection and Skeleton Tracking Using MediaPipe and OpenCV

Introduction:

Human Pose Detection is a computer vision technique that detects and tracks various body parts on images or live video feeds. This process enables computers to analyze and identify human body posture, motion, and alignment via detecting body points called landmarks.

The technology is broadly applied in fitness training, healthcare, gaming, virtual reality, motion analysis, and virtual fitting systems.

Technologies Used:

OpenCV

OpenCV is an open-source library for image and video processing. In this system, it is used for: Webcam video capture, Image frame processing, Output display, Video processing.

It functions as the connector between the webcam and the pose detection program.

MediaPipe

MediaPipe is a Google Machine Learning-based framework for developing real-time perceptual systems.

In MediaPipe, the Pose detection solution recognizes and tracks 33 human body landmarks like: Nose, Eyes, Shoulders, Elbows, Wrists, Hips, Knees, Ankles.

It uses deep learning models that provide accurate estimation of body posture and movements in real time.

Human Pose Detection Process:

Initially, the system activates the webcam to capture consecutive video frames. Each frame acts as an input image that will be fed into the pose detection model.

The obtained image is converted to the RGB format since the MediaPipe Pose model performs faster when analyzing RGB images.

Next, the model processes the image and identifies important body landmarks. These landmarks act as specific points located on the human body.

The landmarks detected by the system have the following features: X-coordinate, Y-coordinate, Visibility score

The process is done in real time while processing each frame captured through the webcam.

Human Pose Skeleton Tracking:

Pose skeleton tracking is realized through connecting detected body landmarks using pose connections.

Automatic lines are drawn between different body parts such as: Shoulder to elbow, Elbow to wrist, Hip to knee, Knee to ankle

The connected landmarks create an artificial human pose skeleton. The skeleton will change its position based on human body motion.

Landmarks on Body Identified

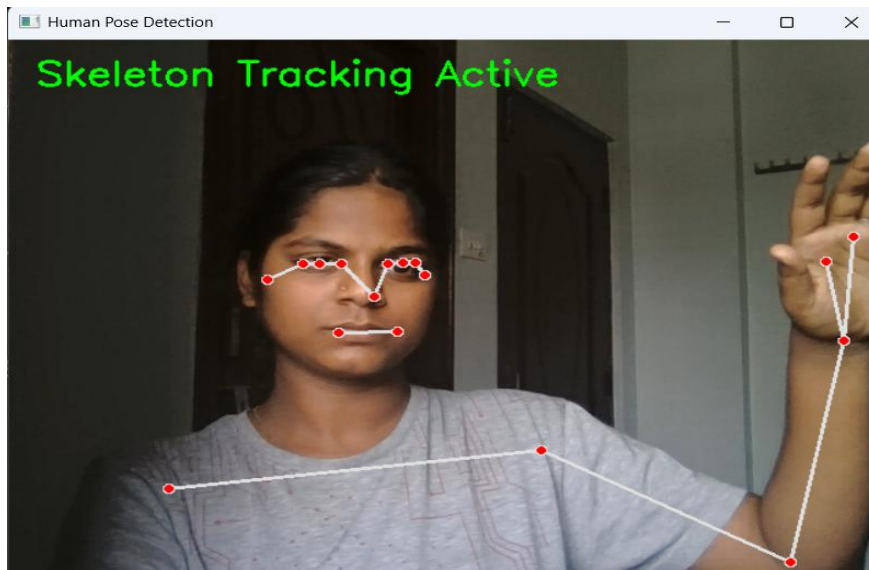
The system is capable of detecting several body landmarks, which include: Head and facial points, Shoulder joints, Arm joints, Hip joints, Leg joints, Foot positions

These landmarks aid in comprehending body posture, orientation, and movement pattern of the user.

CODE:

```
pose_detection.py > ...
1  import cv2
2  import mediapipe as mp
3  mp_pose=mp.solutions.pose
4  pose=mp_pose.Pose()
5  mp_draw=mp.solutions.drawing_utils
6  cap=cv2.VideoCapture(0)
7  while True:
8      success,img=cap.read()
9      if not success:
10         break
11         imgRGB=cv2.cvtColor(img,cv2.COLOR_BGR2RGB)
12         results=pose.process(imgRGB)
13         if results.pose_landmarks:
14             mp_draw.draw_landmarks(
15                 img,
16                 results.pose_landmarks,
17                 mp_pose.POSE_CONNECTIONS
18             )
19             cv2.putText(
20                 img,
21                 "Skeleton Tracking Active",
22                 (20,40),
23                 cv2.FONT_HERSHEY_SIMPLEX,
24                 1,
25                 (0,255,0),
26                 2
27             )
28             cv2.imshow("Human Pose Detection",img)
29             if cv2.waitKey(1)&0xFF==ord('q'):
30                 break
31 cap.release()
32 cv2.destroyAllWindows()
```

Output:



Real-Time Tracking:

All the process of detection and tracking takes place in real time without delay. When a person appears in front of the camera, all the body landmarks and their corresponding skeleton connections are immediately updated.

As a result, this real-time characteristic enables the system to be used for various AI applications and motion tracking devices.

Applications:

Human Pose Detection and Skeleton Tracking have many different applications in daily life, such as: Health and fitness tracking, Yoga position correction, Motion-controlled gaming, Physiotherapy, Animation and motion capturing, Virtual try-ons, Activity and surveillance recognition

Conclusion:

This particular assignment illustrates an example of Human Pose Detection and Skeleton Tracking using the MediaPipe and OpenCV frameworks. The body landmarks of a person were detected in real time and connected to make the skeleton framework virtually.